

```
In [1]: import numpy as np
import pandas as pd
import requests
import re
import json
from bs4 import BeautifulSoup
import dotenv
import os
dotenv.load_dotenv()
```

Out[1]: True

1.A

```
In [2]: expected_columns = 12

def clean_value(value):
    if isinstance(value, str):
        return value.replace('|', '') # Remove '|' character
    return value

df = pd.read_csv("cands22.txt",
                 names=range(expected_columns),
                 converters = {i: clean_value for i in range(expected_columns)},
                 usecols=range(expected_columns))

df.head()
```

Out[2]:

	0	1	2	3	4	5	6	7	8	9	10	11
0	2022	H0AK00105	N00039029	Thomas Lamb (3)	3	AK01					3N	
1	2022	H0AL01055	N00044245	Jerry Carl (R)	R	AL01	AL01	Y	Y	I	RW	
2	2022	H0AL01063	N00044288	Wes Lambert (R)	R	AL01					RN	
3	2022	H0AL01097	N00044750	James Averhart (D)	D	AL01					DN	
4	2022	H0AL02087	N00030768	Martha Roby (R)	R	AL02					RN	

```
In [3]: df.shape
```

Out[3]: (8928, 12)

1.B

```
In [4]: cand = pd.read_csv("cands22.txt", on_bad_lines="skip")
```

```
In [5]: cand.shape
```

Out[5]: (8914, 12)

In [6]: *# This approach simply ignores any lines with an excessive column count (>12) inste*

2

```
In [7]: NASA_KEY = os.getenv("NASA_API_KEY")

def get_useragent():
    url = "https://httpbin.org/user-agent"
    r = requests.get(url)
    useragent = json.loads(r.text)['user-agent']
    return useragent
```

```
In [8]: params = {"start_date": '2024-10-09', 'api_key':NASA_KEY}
headers = {"From": "wfl9zy@virginia.edu", "User-Agent":get_useragent()}
data = requests.get("https://api.nasa.gov/neo/rest/v1/feed", params=params).json()
```

```
In [9]: df_neo_dates = pd.json_normalize(data['near_earth_objects']).columns
```

```
In [10]: neo_dates = pd.json_normalize(data['near_earth_objects']).columns

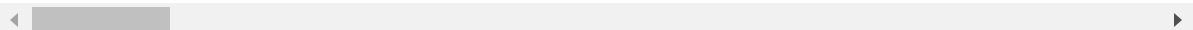
df_neo = pd.DataFrame()
for date in neo_dates:
    df_neo = pd.concat([df_neo, pd.json_normalize(data['near_earth_objects'][date])])
```

```
In [11]: df_neo.head()
```

```
Out[11]:
```

	id	neo_reference_id	name	nasa_jpl_url	absolu
0	2534676	2534676	534676 (2014 VK2)	https://ssd.jpl.nasa.gov/tools/sbdb_lookup.htm...	
1	3826858	3826858	(2018 QE)	https://ssd.jpl.nasa.gov/tools/sbdb_lookup.htm...	
2	3843427	3843427	(2019 OV3)	https://ssd.jpl.nasa.gov/tools/sbdb_lookup.htm...	
3	54105558	54105558	(2021 AG6)	https://ssd.jpl.nasa.gov/tools/sbdb_lookup.htm...	
4	54135427	54135427	(2021 GP3)	https://ssd.jpl.nasa.gov/tools/sbdb_lookup.htm...	

5 rows × 141 columns



3

In [12]: `GENIUS_KEY = os.getenv("GENIUS_API_KEY")`

In [13]: `headers = {'User-Agent': get_useragent(), "From": "wloving77@gmail.com", "Authoriza`

In [14]: `lizzo_id_params = {"q": "Lizzo ID"}

lizzo_data = requests.get("https://api.genius.com/search", headers=headers, params=
lizzo_id = lizzo_data.json()['response']['hits'][0]['result']['primary_artists'][0]`

In [15]: `lizzo_id`

Out[15]: 1424339

In [16]: `lizzo_song_params = {"id": lizzo_id, "sort": "popularity", "per_page": 20}
lizzo_songs = requests.get(f"https://api.genius.com/artists/{lizzo_id}/songs", head`

In [17]: `df_songs = pd.json_normalize(lizzo_songs.json()['response']['songs'])`

In [18]: `df_songs.head()`

Out[18]:

	annotation_count	api_path	artist_names	full_title	
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0	13	/songs/5923313	Lizzy McAlpine	ceilings by Lizzy McAlpine	https://images.geniu
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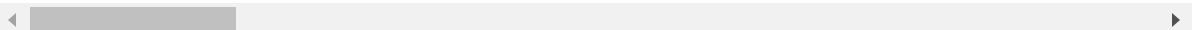
1	21	/songs/7327417	Lizzy McAlpine	doomsday by Lizzy McAlpine	https://images.genius.c
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2	2	/songs/5721612	Lizzy McAlpine	Ruined (Demo) by Lizzy McAlpine	https://images.genius.c
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3	13	/songs/7610877	Lizzy McAlpine	chemtrails by Lizzy McAlpine	https://images.genius.
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4	6	/songs/10009375	Lizzy McAlpine	Older by Lizzy McAlpine	https://images.genius.c
---	---	-----------------	----------------	-------------------------	-------------------------

5 rows × 37 columns



4

```
In [19]: # The general / endpoint on Genius.com is not disallowed in https://genius.com/robo
```

```
In [20]: # returns cleaned up Lyrics from a Genius.com endpoint
def get_clean_lyrics(url):

    response = requests.get(url)
    soup1 = BeautifulSoup(response.text)

    lyrics_div = str(soup1.find('div', class_='Lyrics__Container-sc-1ynbvzw-1'))

    soup2 = BeautifulSoup(lyrics_div, 'html.parser')
    for br in soup2.find_all("br"):
        br.replace_with("\n")

    text = soup2.get_text(separator="\n")
    text = re.sub(r'\n+', '\n', text).strip()
    return text
```

```
In [21]: lyrics_goodAsHell = get_clean_lyrics("https://genius.com/Lizzo-good-as-hell-lyrics")
```

```
In [22]: print(lyrics_goodAsHell)
```

```
[Chorus]
I do my hair toss, check my nails
Baby, how you feelin'? (Feelin' good as hell)
Hair toss, check my nails
Baby, how you feelin'? (Feelin' good as hell)
[Verse 1]
Woo, child, tired of the bullshit
Go on, dust your shoulders off, keep it moving
Yes, Lord, tryna get some new shit
In there, swimwear, going-to-the-pool shit
Come now, come dry your eyes
You know you a star, you can touch the sky
I know that it's hard, but you have to try
If you need advice, let me simplify
[Pre-Chorus]
If he don't love you anymore
Just walk your fine ass out the door
[Chorus]
I do my hair toss, check my nails
Baby, how you feelin'? (Feelin' good as hell)
Hair toss, check my nails
Baby, how you feelin'? (Feelin' good as hell)
(Feeling good as hell)
Baby, how you feelin'? (Feelin' good as hell)
```

```
In [23]: def extract_urls(url):
    response = requests.get(url)

    soup1 = BeautifulSoup(response.text)
```

```

track_list = str(soup1.find("ol", class_="AlbumTracklist__Container-sc-123giuo-

# Parse the HTML content
soup2 = BeautifulSoup(track_list, 'html.parser')

# Find all 'a' tags and extract the 'href' attribute
urls = [a['href'] for a in soup2.find_all('a', href=True)]

return urls

```

```
In [24]: urls = extract_urls("https://genius.com/Lizzo-good-as-hell-lyrics")
```

```
In [25]: lyrics_for_album = []
        for url in urls:
            lyrics_for_album.append(get_clean_lyrics(url))
```

```
In [26]: print(lyrics_for_album[0])
```

```

[Intro]
I'm cryin' 'cause I love you, oh
(Ya-ya-ya, ya-ya, ya-ya)
[Verse 1]
Never been in love before
What the fuck are fucking feelings, yo?
Once upon a time, I was a ho
I don't even wanna ho no mo'
Got you something from the liquor store
Little bit of Lizzo and some Mo
Tryna open up a little mo'
Sorry if my heart a little slow
[Chorus]
I thought that I didn't care
I thought I was love-impaired
But baby, baby
I don't know what I'm gon' do
I'm cryin' 'cause I love you, oh
Yes, you (Ya-ya, ya-ya)
[Verse 2]
Got me standing in the rain
Gotta get my hair pressed again
I would do it for you all, my friend
Ready, baby? Will you be my man?
Wanna put you on a plane
Fly you out to wherever I am
Catch you on the low, I was ashamed
Now I'm crazy, 'bout to tat your name

```

```
In [27]: print(lyrics_for_album[3])
```

[Intro]
 True story
 No glory, let's go
 [Verse 1]
 Yeah, the old me used to love a Gemini
 Like a threesome, fuckin' with him every night
 A lotta two-faced people show me both sides
 So I figured out I gotta be my own type
 [Pre-Chorus 1]
 They used to say to get a man, you had to know how to look
 They used to say to keep a man, you had to know how to cook
 But I'm solo in Soho, sippin' Soju in Malibu
 It's a me, myself kinda attitude
 [Chorus]
 'Cause I'm my own soulmate (Yeah, yeah)
 I know how to love me (Love me)
 I know that I'm always gonna hold me down
 Yeah, I'm my own soulmate (Yeah, yeah)
 No, I'm never lonely (Lonely)
 I know I'm a queen, but I don't need no crown
 Look up in the mirror like "Damn, she the one"
 [Post-Chorus]
 One, one, one, one, one, one (The one)
 Like "Damn, she the one"
 One, one, one (The one)
 Bad bitch in the mirror like "Yeah, I'm in love"
 Love, love, love, love, love, love (In love)
 Like "Yeah, I'm in love"
 Love, love, love (In love)
 Look up in the mirror like "Damn, she the one"

5

```
In [28]: root = "https://maps.spirithalloween.com"
         endpoint = "/api/getAsyncLocations"

         headers = {"From": "wf19zy@virginia.edu", "User-Agent": get_useragent()}
         parameters = {"template": "domain", "level": "domain", 'lat': 40.380028, 'lng': -97.91015}

In [29]: halloween_mapdata = requests.get(root+endpoint, headers=headers, params=parameters)

In [30]: halloween_stores = halloween_mapdata.json()['markers']

In [31]: info_list = []
         for i in range(len(halloween_stores)):
             info_list.append(halloween_stores[i]['info'])
```

```
In [32]: def get_all_former_locations(info_list):
        former_locations = []
        for i in range(len(info_list)):
            soup = BeautifulSoup(info_list[i])
            former_locations.append(soup.find("div", class_="address-two").get_text(strip=True))
        return former_locations
```

```
In [33]: all_former_locations = get_all_former_locations(info_list)
```

```
In [34]: series = pd.Series(all_former_locations)
```

```
In [35]: series.value_counts()
```

```
Out[35]: Former Rite Aid          71
        Former Tuesday Morning    62
        Former Sears              53
        Former Bed Bath & Beyond  49
        Former CVS                42
        ..
        Former Luxury Home Furniture  1
        Located between Walmart and Meijers  1
        Next to Guitar Center        1
        Next to Sky Zone Trampoline Park  1
        Across from the Food Court    1
        Name: count, Length: 749, dtype: int64
```

```
In [ ]:
```